

```
import numpy as np
```

✓ SESSION 4 — Array Operations

Array operations are the core strength of NumPy.

Why do we need array operations?

Because in scientific computing and machine learning we frequently perform:

- Element-wise arithmetic
- Comparisons
- Logical filtering
- Mathematical transformations

NumPy allows these operations without explicit loops. This is called vectorization.

1. Arithmetic Operations

Operators supported:

- + Addition
- - Subtraction
- * Multiplication
- / Division
- // Floor Division
- ** Power
- % Modulus

These operate element-wise when applied to arrays.

```
import numpy as np

a = np.array([10, 20, 30])
b = np.array([2, 4, 5])

print("Addition:", a + b)
print("Subtraction:", a - b)
print("Multiplication:", a * b)
print("Division:", a / b)
print("Floor Division:", a // b)
print("Power:", a ** 2)
print("Modulus:", a % b)
```

```
Addition: [12 24 35]
Subtraction: [ 8 16 25]
Multiplication: [ 20 80 150]
Division: [5. 5. 6.]
Floor Division: [5 5 6]
Power: [100 400 900]
Modulus: [0 0 0]
```

```
A[2]
```

Why these operators exist:

- Enable vectorized computation
- Avoid explicit loops
- Memory efficient

2. Comparison Operators

Used for element-wise comparison.

`==` Equal

`!=` Not equal

Greater than

`<` Less than

`=` Greater than or equal

`<=` Less than or equal

Output is a Boolean array.

```
a = np.array([5, 10, 15])

print("Greater than 8:", a < 8)# Returns Boolean
print("Greater than 8:", a[a < 8])# Returns Value's that satisfies the condition
print("Equal to 10:", a == 10)# Returns Boolean

[ True False False False]
[False  True  True  True]
[ True  True  True False]
Greater than 8: [5]
Equal to 10: [False False False False]
```

These Boolean arrays are used for filtering.

```
print("Elements greater than 8:", a[a > 8])

Elements greater than 8: [10 15]
```

3. Logical Operations

Used when combining multiple conditions.

Functions:

`np.logical_and(condition1, condition2)`

`np.logical_or(condition1, condition2)`

`np.logical_not(condition)`

Why these exist:

- Combine Boolean conditions
- Enable complex filtering
- Used heavily in data analysis

```
a = np.array([5, 8, 15, 20])

cond1 = a > 8
cond2 = a < 18
```

```
print("Logical AND:", np.logical_and(cond1, cond2))
print("Filtered Result:", a[np.logical_and(cond1, cond2)])

print("Logical OR:", np.logical_or(cond1, cond2))
print("Logical OR:", a[np.logical_or(cond1, cond2)])

print("Logical NOT:", np.logical_not(cond1))
print("Logical NOT:", a[np.logical_not(cond1)])
```

```
Logical AND: [False False  True False]
Filtered Result: [15]
Logical OR: [ True  True  True  True]
Logical OR: [ 5  8 15 20]
Logical NOT: [ True  True False False]
Logical NOT: [5 8]
```

4. Broadcasting

Broadcasting allows operations between arrays of different shapes.

Rule: NumPy automatically expands dimensions when compatible.

Example:

```
a = np.array([[1, 2, 3],
              [4, 5, 6]])

b = np.array([10, 20, 30])

print("Broadcasted Addition:\n", a + b)
```

```
Broadcasted Addition:
[[11 22 33]
 [14 25 36]]
```

Why Broadcasting is Powerful:

- Avoids manual reshaping
- Reduces memory duplication
- Speeds up computations

5. Element-wise vs Aggregate Operations

Element-wise: Operation applied to each element individually.

Aggregate: Operation reduces array to single value.

```
arr = np.array([1, 2, 3, 4])

print("Element-wise square:", arr ** 2)
print("Aggregate sum:", np.sum(arr))
```

```
Element-wise square: [ 1  4  9 16]
Aggregate sum: 10
```

SESSION 5 — Mathematical & Statistical Functions

NumPy provides optimized mathematical functions.

Why separate functions exist when operators already work?

Because:

- Functions allow explicit usage
- Useful in functional programming
- More readable in complex expressions
- Support broadcasting and advanced parameters

✓ 1. Basic Mathematical Functions

Functions:

- np.add()
- np.subtract()
- np.multiply()
- np.divide()
- np.power()
- np.mod()
- np.sqrt()
- np.exp()
- np.log()
- np.log10()

```
a = np.array([4, 9, 16])
# a = np.array([1, 2, 3])
b = np.array([2, 3, 4])
c = np.array([2, 8, 4])

print("Add:", np.add(a, b))
z = np.add(a, b)
print(np.add(z, c))
print("Subtract:", np.subtract(a, b))
print("Multiply:", np.multiply(a, b))
print("Divide:", np.divide(a, b))
print("Power:", np.power(a, 2))
print("Mod:", np.mod(a, b))

print("Square root:", np.sqrt(a))
print("Exponential:", np.exp(a))
print("Natural log:", np.log(a))
print("Log base 10:", np.log10(a))
```

```
Add: [ 6 12 20]
[ 8 20 24]
Subtract: [ 2  6 12]
Multiply: [ 8 27 64]
Divide: [2.  3.  4.]
Power: [ 16  81 256]
Mod: [0 0 0]
Square root: [2.  3.  4.]
Exponential: [5.45981500e+01 8.10308393e+03 8.88611052e+06]
Natural log: [1.38629436 2.19722458 2.77258872]
Log base 10: [0.60205999 0.95424251 1.20411998]
```

Functions:

np.round()
np.floor()
np.ceil()

Used in financial calculations and data cleaning.

```
arr = np.array([1.4, 2.6, 3.2])

print("Round:", np.round(arr))
print("Floor:", np.floor(arr))
print("Ceil:", np.ceil(arr))
```

```
Round: [1.  3.  3.]
Floor: [1.  2.  3.]
Ceil:  [2.  3.  4.]
```

Statistical operations reduce arrays.

Functions:

```
np.mean()
np.median()
np.std()
np.var()
np.min()
np.max()
np.sum()
np.cumsum()
np.prod()
np.percentile()
```

Why these exist:

- Data analysis
- Machine learning preprocessing
- Performance metrics

```
data = np.array([10, 24, 30, 36, 50])
# data = np.array([1, 1])

print("Mean:", np.mean(data))
print("Median:", np.median(data))
print("Standard Deviation:", np.std(data))
print("Variance:", np.var(data))
print("Minimum:", np.min(data))
print("Maximum:", np.max(data))
print("Sum:", np.sum(data))
print("Cumulative Sum:", np.cumsum(data))
print("Product:", np.prod(data))
print("90th Percentile:", np.percentile(data, 90))
```

```
Mean: 30.0
Median: 30.0
Standard Deviation: 13.206059215375342
Variance: 174.4
Minimum: 10
Maximum: 50
Sum: 150
Cumulative Sum: [ 10  34  64 100 150]
Product: 12960000
90th Percentile: 44.4
```

✓ NumPy Practice Set — Session 4 & 5

(Array Operations + Mathematical & Statistical Functions)

EASY LEVEL

1. Create two arrays: `a = [5, 10, 15]` `b = [1, 2, 3]` Perform element-wise addition and multiplication.
2. Given array: `arr = [4, 8, 12, 16]` Extract elements greater than 10.
3. Create a 2×3 array of ones and add 5 to every element using broadcasting.
4. Given array: `arr = [10, 20, 30, 40]` Compute:
 - Mean
 - Sum
 - Maximum

MEDIUM LEVEL

6. Create two arrays: `a = [2, 4, 6, 8]` `b = [1, 2, 3, 4]` Perform floor division and modulus operations.
7. Generate a 3×3 random integer matrix between 1 and 20. Find its standard deviation and variance.
8. Create an array from 1 to 9 and reshape it into 3×3 . Compute cumulative sum (`cumsum`).
9. Create array: `arr = [-3, -1, 0, 1, 3]` Apply:
 - Square
 - Square root (only valid values)
10. Using logical operations: Given array: `arr = [5, 10, 15, 20, 25]` Extract values between 10 and 20 inclusive.

HARD LEVEL

12. Generate 1000 random numbers from normal distribution. Compute:
 - Mean
 - Standard deviation
 - 95th percentile
13. Create two matrices: A (2×3) and B (3×2). Transpose A and verify its shape.
14. Create array: `arr = [1, 2, 3, 4]` Compute:
 - Exponential of each element
 - Natural log
 - Log base 10
15. Create array: `arr = [1.2, 2.5, 3.7, 4.1]` Apply:
 - `round()`
 - `floor()`
 - `ceil()` Explain differences based on output.

```
import numpy as np

# -----
# EASY LEVEL SOLUTIONS
# -----

# 1
a = np.array([5, 10, 15])
b = np.array([1, 2, 3])

print("Q1 Addition:", a + b)      # Element-wise addition
print("Q1 Multiplication:", a * b) # Element-wise multiplication
```

```
# 2
arr = np.array([4, 8, 12, 16])
print("Q2 Elements > 10:", arr[arr > 10]) # Boolean filtering

# 3
arr = np.ones((2, 3))
print("Q3 Broadcasting Result:\n", arr + 5) # Scalar broadcasting

# 4
arr = np.array([10, 20, 30, 40])
print("Q4 Mean:", np.mean(arr))
print("Q4 Sum:", np.sum(arr))
print("Q4 Max:", np.max(arr))

# 5
angle_rad = np.radians(45) # Convert degrees to radians
print("Q5 Sine of 45 degrees:", np.sin(angle_rad))

# -----
# MEDIUM LEVEL SOLUTIONS
# -----

# 6
a = np.array([2, 4, 6, 8])
b = np.array([1, 2, 3, 4])
print("Q6 Floor Division:", a // b)
print("Q6 Modulus:", a % b)

# 7
np.random.seed(0)
matrix = np.random.randint(1, 21, (3, 3))
print("Q7 Matrix:\n", matrix)
print("Q7 Std:", np.std(matrix))
print("Q7 Variance:", np.var(matrix))

# 8
arr = np.arange(1, 10).reshape(3, 3)
print("Q8 Matrix:\n", arr)
print("Q8 Cumulative Sum:", np.cumsum(arr))

# 9
arr = np.array([-3, -1, 0, 1, 3])
print("Q9 Square:", np.power(arr, 2))

# Square root only valid for non-negative numbers
print("Q9 Square Root (valid only):", np.sqrt(arr[arr >= 0]))

# 10
arr = np.array([5, 10, 15, 20, 25])
condition = np.logical_and(arr >= 10, arr <= 20)
print("Q10 Between 10 and 20:", arr[condition])

# -----
# HARD LEVEL SOLUTIONS
# -----

# 11
matrix = np.arange(12).reshape(3, 4)
```

```

vector = np.array([10, 20, 30, 40])
print("Q11 Original Matrix:\n", matrix)
print("Q11 After Broadcasting:\n", matrix + vector)

# 12
np.random.seed(1)
data = np.random.randn(1000)
print("Q12 Mean:", np.mean(data))
print("Q12 Std:", np.std(data))
print("Q12 95th Percentile:", np.percentile(data, 95))

# 13
A = np.arange(6).reshape(2, 3)
B = np.arange(6).reshape(3, 2)
print("Q13 A:\n", A)
print("Q13 Transposed A:\n", A.T)
print("Q13 Shape of A.T:", A.T.shape)

# 14
arr = np.array([1, 2, 3, 4])
print("Q14 Exponential:", np.exp(arr))
print("Q14 Natural Log:", np.log(arr))
print("Q14 Log Base 10:", np.log10(arr))

# 15
arr = np.array([1.2, 2.5, 3.7, 4.1])
print("Q15 Round:", np.round(arr))      # Nearest integer
print("Q15 Floor:", np.floor(arr))      # Round down
print("Q15 Ceil:", np.ceil(arr))        # Round up

```

```

Q1 Addition: [ 6 12 18]
Q1 Multiplication: [ 5 20 45]
Q2 Elements > 10: [12 16]
Q3 Broadcasting Result:
[[6. 6. 6.]
 [6. 6. 6.]]
Q4 Mean: 25.0
Q4 Sum: 100
Q4 Max: 40
Q5 Sine of 45 degrees: 0.7071067811865475
Q6 Floor Division: [2 2 2 2]
Q6 Modulus: [0 0 0 0]
Q7 Matrix:
[[13 16  1]
 [ 4  4  8]
 [10 20 19]]
Q7 Std: 6.499762578759852
Q7 Variance: 42.24691358024692
Q8 Matrix:
[[1 2 3]
 [4 5 6]
 [7 8 9]]
Q8 Cumulative Sum: [ 1  3  6 10 15 21 28 36 45]
Q9 Square: [9 1 0 1 9]
Q9 Square Root (valid only): [0.          1.          1.73205081]
Q10 Between 10 and 20: [10 15 20]
Q11 Original Matrix:
[[ 0  1  2  3]
 [ 4  5  6  7]
 [ 8  9 10 11]]
Q11 After Broadcasting:
[[10 21 32 43]
 [14 25 36 47]
 [18 29 40 51]]
Q12 Mean: 0.03881247615960185
Q12 Std: 0.9810041339322116
Q12 95th Percentile: 1.6278527911730007

```

```

Q13 A:
[[0 1 2]
 [3 4 5]]
Q13 Transposed A:
[[0 3]
 [1 4]
 [2 5]]
Q13 Shape of A.T: (3, 2)
Q14 Exponential: [ 2.71828183  7.3890561  20.08553692  54.59815003]
Q14 Natural Log: [0.          0.69314718  1.09861229  1.38629436]
Q14 Log Base 10: [0.          0.30103    0.47712125  0.60205999]
Q15 Round: [1.  2.  4.  4.]
Q15 Floor: [1.  2.  3.  4.]
Q15 Ceil: [2.  3.  4.  5.]

```

✓ SESSION 6 — Indexing & Slicing in NumPy

Indexing and slicing are fundamental for:

- Accessing specific elements
- Extracting subsets of data
- Filtering datasets
- Modifying portions of arrays

Without indexing and slicing, data analysis is not practical.

1. Basic Indexing (1D Arrays)

Why It Exists

To access individual elements from an array.

Syntax

```
arr[index]
```

Index starts from 0.

```

import numpy as np

arr = np.array([10, 20, 30, 40])

print("Element at index 0:", arr[0])
print("Element at index 2:", arr[2])

```

```

Element at index 0: 10
Element at index 2: 30

```

✓ 2. Negative Indexing

Why It Exists

To access elements from the end of the array.

Syntax

```
arr[-1] → last element
```

```
arr[-2] → second last element
```

```
print("Last element:", arr[-1])
print("Second last:", arr[-2])
```

```
Last element: 40
Second last: 30
```

3. Basic Indexing (2D Arrays)

Why It Exists

To access elements in matrices.

Syntax

`arr[row_index, column_index]`

```
matrix = np.array([[1, 2, 3],
                   [4, 5, 6],
                   [7, 8, 9]])

print("Element at (2,3):", matrix[1, 2])
```

```
Element at (1,2): 6
```

4. Slicing

Why It Exists

To extract subarrays.

Syntax

`arr[start:stop:step]`

- start → inclusive
- stop → exclusive
- step → interval

```
arr = np.array([0, 10, 20, 30, 40, 50])

print("Slice 1:4:", arr[1:4])
print("Slice with step 2:", arr[0:6:2])
```

```
Slice 1:4: [10 20 30]
Slice with step 2: [ 0 20 40]
```

5. 2D Slicing

Syntax:

`arr[row_start:row_end, col_start:col_end]`

```
matrix = np.array([[1, 2, 3],
                   [4, 5, 6],
                   [7, 8, 9]])

print("Submatrix:\n", matrix[0:2, 1:3])
```

```
Submatrix:
[[2 3]
 [5 6]]
```

✓ 7. Boolean Indexing

Why It Exists

To filter arrays based on conditions.

Syntax

`arr[condition]`

```
arr = np.array([5, 10, 15, 20])

print("Elements > 10:", arr > 10)
print("Elements > 10:", arr[arr > 10])

Elements > 10: [False False  True  True]
Elements > 10: [15 20]
```

✓ 8. Indexing with np.where()

Why It Exists

To get indices where condition is True.

Syntax

`np.where(condition)`

```
arr = np.array([5, 10, 15, 20])

indices = np.where(arr > 11)
print("Indices:", indices)
print("Values:", arr[indices])
```

```
Indices: (array([2, 3]),)
Values: [15 20]
```

```
arr = np.array([5, 10, 15, 20])

# print(np.where(arr > 10))
print(arr[np.where(arr > 10)])
# print("Indices:", indices)
# print("Values:", arr[indices])
```

```
[15 20]
```

✓ Practice Questions — Session 6 (Indexing & Slicing)

SESSION 6

1.

Create a 1D NumPy array: `arr = [10, 20, 30, 40, 50]`

Print:

- First element
 - Last element using negative indexing
-

2.

Given array: `arr = [5, 10, 15, 20, 25, 30]`

Extract elements from index 1 to 4 using slicing.

3.

Given array: `arr = [0, 2, 4, 6, 8, 10]`

Extract every second element using step slicing.

4.

Create a 2D array: `[[1, 2, 3], [4, 5, 6], [7, 8, 9]]`

Print the element located at: Row 2, Column 1 (using proper indexing).

5.

Using the same 2D array from Question 4, Extract the first two rows.

6.

Using the same 2D array, Extract the submatrix: `[[2, 3], [5, 6]]`

7.

Create array: `arr = [10, 20, 30, 40, 50]`

Using fancy indexing, extract:

- 1st element
 - 3rd element
 - 5th element
-

8.

Create array: `arr = [5, 10, 15, 20, 25]`

Using boolean indexing, extract elements greater than 12.

9.

Given array: `arr = [3, 6, 9, 12, 15]`

Use `np.where()` to find indices where values are greater than 8.

10.

Create array: `arr = [1, 2, 3, 4, 5]`

Reverse the array using slicing.

REVISION QUESTIONS (Previous Sessions — 2 Questions)

11. (Array Operations)

Create arrays: `a = [2, 4, 6]` `b = [1, 2, 3]`

Perform:

- Element-wise multiplication
 - Element-wise power (a squared)
-

12. (Mathematical Functions)

Create array: `arr = [4, 9, 16, 25]`

Compute:

- Square root
- Mean
- Maximum value

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